# Conditional Statements
(Assignment Solutions)

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**Question 1** :

```cpp
int main() {
   int num;
   cin >> num;

   if (num > 0) {
     cout << "number is positive.\n";
   } else if (num < 0) {
     cout << "number is negative.\n";
   } else {
     cout << "number is zero.\n";
   }

   return 0;
}
```

**Question 2** :

```cpp
int main() {
   int year;
   cin >> year;

   if (year % 400 == 0) {
     cout << year << " is a leap year.\n";
   } else if (year % 100 == 0) {
     cout << year << " is NOT a leap year.\n";
   } else if (year % 4 == 0) {
     cout << year << " is a leap year.\n";
   } else {
     cout << year << " is NOT a leap year.\n";
   }

   return 0;
}
```

**Question 3** : x = 0(false) & y = 63

**Question 4** : Output will be “Bye”.

**Question 5** :

```cpp
int main() {
   int n;
   cout << "enter a 3 digit number : ";
   cin >> n;

   int num = n; //creating a copy

   int dig1 = num % 10;
   num /= 10;
   int dig2 = num % 10;
   num /= 10;
   int dig3 = num;

   int cubeSum = dig1*dig1*dig1 + dig2*dig2*dig2 + dig3*dig3*dig3;

   if (cubeSum == n) {
       cout << n << " is an Armstrong Number\n";
   } else {
       cout << n << " is NOT an Armstrong Number\n";
   }

   return 0;
}
```

**Question 6** :

***typedef*** keyword in C++ is used for aliasing existing data types, user-defined data types, and pointers to a more meaningful name.

Eg : *typedef int myInt*

This allows us to declare new integers using myInt, instead of int.

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***Macros*** can be of any type. Macros can even be any code block containing statements, loops, function calls etc. They are expanded by the preprocessor before compilation takes place.

***const*** keyword is used to define the constant value that cannot change during program execution.

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